

# Sohel Ahmed

Senior Research Fellow

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🌐 ResearchGate Profile



## Professional Summary

PhD Research Scholar in Applied Mathematics at Cooch Behar Panchanan Barma University, working in biofluid mechanics, solute dispersion, and drug delivery modeling using analytical and numerical approaches. Skilled in mathematical modeling, partial differential equations, numerical simulation, and scientific computing with MATLAB, Python, Mathematica, and  $\text{\LaTeX}$ . Qualified Joint CSIR-UGC NET with AIR 160, GATE, and recipient of CSIR fellowship for doctoral research. Interested in applying mathematical reasoning, computational methods, and AI-assisted research tools to solve complex scientific and analytical problems.

## Education

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|----------------------------------------------------------------------------|---------------------------------|
| <b>PhD in Mathematics</b>                                                  | <b>Nov 2022 – Expected 2027</b> |
| Cooch Behar Panchanan Barma University, West Bengal, India                 |                                 |
| Research Area: Biofluid Mechanics, Solute Dispersion, Drug Delivery        |                                 |
| <b>M.Sc. in Mathematics</b>                                                | <b>2020</b>                     |
| Cooch Behar Panchanan Barma University                                     |                                 |
| Specialization: Applied Mathematics — Fluid Mechanics and Plasma Mechanics |                                 |
| <b>B.Sc. in Mathematics (Honours)</b>                                      | <b>2018</b>                     |
| Dinhata College, Cooch Behar Panchanan Barma University                    |                                 |

## Technical Skills

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|------------------------------|------------------------------------------------------------------------------------------------|
| <b>Programming Languages</b> | Python, MATLAB, Mathematica                                                                    |
| <b>Scientific Writing</b>    | $\text{\LaTeX}$ , Academic Writing, Research Documentation                                     |
| <b>Core Mathematics</b>      | Partial Differential Equations, Fluid Mechanics, Mathematical Modeling                         |
| <b>Computational Methods</b> | Numerical Simulation, Scientific Computing, Analytical and Numerical Methods                   |
| <b>Research Interests</b>    | Biofluid Mechanics, Solute Dispersion, Drug Delivery Modeling, AI-assisted Scientific Research |

## Research Experience

### Doctoral Researcher — Biofluid Mechanics and Solute Dispersion

Department of Mathematics, Cooch Behar Panchanan Barma University

- Conducting theoretical research on solute dispersion and drug delivery systems using analytical and numerical techniques.
- Working with mathematical models governed by differential equations arising in biofluid transport phenomena.
- Developing research manuscripts involving model formulation, solution methodology, parameter analysis, and scientific interpretation.

## Teaching Experience

### Teaching Assistant

Cooch Behar Panchanan Barma University

- Taught Topology for two semesters in the Department of Mathematics.

- Guided students in understanding abstract mathematical structures, proofs, and problem-solving methods.
- Conducted classroom discussions and supported academic learning through examples and explanations.

## **Publications and Research Work**

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- Published one book chapter in Applied Mathematics.
- Preparing research manuscripts for communication to peer-reviewed journals.

## **Achievements and Certifications**

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- Qualified Joint CSIR-UGC NET with All India Rank 160.
- Recipient of CSIR Fellowship for pursuing doctoral research.
- Qualified GATE Examination, 2022.
- Completed 12-week NPTEL Python course organized by IIT Madras.
- Participated in Hacktoberfest 2021 and 2022.

## **Selected Academic Project**

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### **A Brief Report on Free Convection from a Heated Vertical Plate**

- Studied heat transfer and fluid flow behavior using concepts from fluid mechanics and differential equations.
- Applied mathematical techniques to understand free convection phenomena.

## **Languages**

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English, Bengali, Hindi